

YUHENG DING

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EDUCATION

Carnegie Mellon University

Master of Science in Intelligent Information Systems (ML/NLP @ SCS LTI)

Pittsburgh, PA

December 2026

University of California, Los Angeles (UCLA)

Bachelor of Science in Computer Science, Bachelor of Science in Applied Mathematics

Los Angeles, CA

June 2025

GPA: 3.88/4.0, Cum Laude

Relevant Courses: Algorithm and Complexity, Database Systems, Data Structure, Object Oriented Design, Operating Systems, Computer Network, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing

EXPERIENCE

VisionX LLC

Full-Stack Software Engineer Intern

San Jose, CA (Remote)

July 2023 - October 2023

- Collaborated with a team of four engineers and contributed to software development of a full-stack e-commerce platform, with a focus on promoting a healthy lifestyle and fostering environmental sustainability.
- Engineered a product page UI template using **Python Django** and **MongoDB**, empowering automatic page creation from database; enhanced scalability and amplified product creation efficiency by 400% over manual page inputting.
- Implemented user information gathering, payment workflow with Stripe API, and email order confirmations; achieved a 30% reduction in checkout process response time.
- Coordinated with a colleague to enhance online consultation reservation system, integrating voice and video chat functionalities applying **WebSocket** and **WebRTC** technologies, resulting in 60% decrease in confirmation time.

RESEARCH

UCLA-NLP | Mentor: Kai-Wei Chang, Nanyun Peng

Undergrad Researcher

Los Angeles, CA

June 2024 - June 2025

- Developed a comprehensive benchmark for evaluating visual language models (VLMs) on perception and reasoning quality in QA tasks; tested on 26 VLMs to assess self-critique and correction abilities (**CVPR 2025, Accepted [Link](#)**).
- Managed a team of 3 annotators and built a **Python**-based annotation tool, allowing efficient curation of over 1,600 fine-grained annotated samples.
- Introduced a novel evaluation metric, VISCore, combining answer accuracy, stepwise critique accuracy, and critique quality to yield more accurate model assessments.
- Implemented an agent-based critique generation method at each reasoning step, improving critique accuracy by 13.5%.

PROJECTS

High Performance URL Shortening Web Server - Tech Lead & Contributor

April 2025 - June 2025

- Led a team of 4 in building a low-cost, high-performance async TCP server in **C++** with **Boost.Asio** following test-driven development; authored a system design document and technical documentation to guide product architecture.
- Designed application architecture design for a scalable URL shortening service with **Redis** caching and **PostgreSQL** persistence, leveraging connection pooling to support 1,000+ concurrent users.
- Deployed service on **Google Cloud Platform** using **Docker** and Compute Engine; automated CI/CD with **Cloud Build**, to ensure zero-downtime deployments and automatic recovery through automated testing and health checks.

Job-Search - Independent Developer

October 2023 - January 2024

- Initiated and constructed a job search platform, personalizing job recommendations merely based on user views and likes, incorporated Search, Favorite, and Recommendation functionalities in **Java Servlets** with **REST APIs**.
- Designed a recommendation engine with dense-embeddings and **AI**-based keyword extraction to model user interests.
- Implemented **Redis** as a high-speed vector database to store job/user embeddings and precomputed results, reducing embedding API calls by 80%+ and enabling sub-millisecond cosine-similarity retrieval.
- Deployed on **AWS EC2** virtual machine, combining **Redis** and **Amazon RDS** to cache and persist search and favorite related data, reducing latency by 90%.

SKILLS

Programming Languages: Java, Python, C++, C, SQL, JavaScript, Bash

Frameworks & Libraries: Spark, React, HTML, CSS, Django, Spring Boot, Tomcat, PyTorch, NumPy

Databases & Data Stores: PostgreSQL, MySQL, SQLite, Redis, MongoDB

Cloud Platforms: AWS, Google Cloud Platform (GCP), Microsoft Azure

DevOps & Tools: Docker, Kubernetes, Git, Linux